

CHE 435/525 *Process Systems Analysis and Control*

Lecture 11: **Frequency Response and Bode diagram**

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Recapture (Lecture 7–10): Time-domain analysis

- **Step, impulse (, ramp) responses**
 - Characteristics of **first- and second-order systems**
 - **Poles, zeros, gain, time delay** and their implications
 - Standard gain/time constant forms of transfer functions
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- Transfer function model (Lecture 6) from linearized ODEs
 - Qualitative behavior analysis (Lecture 5) based on L.T. (Lecture 4)

Learning objectives of this lecture

- ① Explain the concepts of **amplitude ratio** and **phase angle** of $G(s)$
- ② Make **Bode diagrams** to represent the frequency response
- ③ Calculate frequency response of **first-order lag, lead-lag, integrating, delayed** systems

Frequency Response

- Suppose that the input is sinusoidal: $u(t) = A \sin(\omega t)$
 - ▶ Radial frequency: ω [rad/min (or sec)]
 - ▶ Period: $T = 2\pi/\omega$, frequency: $f = 1/T = \omega/2\pi$ [cycles/min(or sec)]
 - ▶ Amplitude: $A > 0$
- Frequency response: $y(t) = \hat{A} \sin(\omega t + \phi)$ will hold, when t is sufficiently large
 - ▶ Output will also be sinusoidal, with the same period (frequency)
 - ▶ It is of interest to know $\hat{A}/A$ (amplitude ratio) and ϕ (phase angle)
- Why does this matter?
 - ▶ Process disturbances can be viewed as small oscillations
 - ▶ Frequency response **will** play an important role in control
 - ▶ Applied in relevant engineering areas – communications, experimental data processing, AI, etc.

**Testimony of ChatGPT

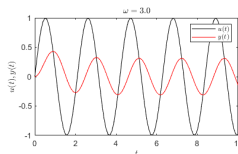
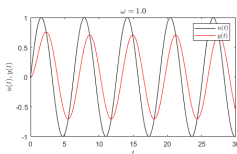
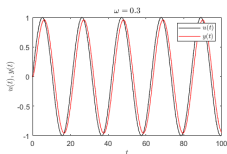
Dr. Tang asks: Some people believe that frequency response should not be taught to undergraduates because it's too hard. Do you agree?

I don't agree that frequency response should be excluded from undergraduate curricula. While it can be challenging, it offers valuable insights that are essential for a comprehensive understanding of control systems. Here are a few reasons.

- 1 Fundamental Concept: Frequency response is foundational to control theory. It connects theory with practical applications like filter design, communications, ...
- 2 Visualization of System Behavior: Tools like Bode plots and Nyquist diagrams provide intuitive ways to understand system behavior, stability, and performance **without complex mathematics**.
- 3 Industry Relevance: Frequency response techniques are **widely used in industries** ranging from aerospace to automotive and electronics. Exposing students to these concepts better prepares them for real-world challenges.
- 4 Supports Robust Design: Frequency response aids in designing systems that are robust to disturbances and uncertainties, an essential skill in engineering practice.
- 5 Enhances Problem-Solving Skills: Learning challenging topics like frequency response helps students develop critical thinking and problem-solving skills, which are valuable beyond control theory.

Simulation: First-order lag

- Consider $G(s) = \frac{1}{s+1}$, $u(t) = \sin(\omega t)$
 $t = 0:0.1:(30/\omega)$; $u = \sin(\omega t)$;
 $y = \text{lsim}(\text{tf}(1, [1, 1]), u, t)$;



- Observation: $\hat{A}/A$ and ϕ depends on ω

Analysis: First-order lag

- Consider $G(s) = \frac{K}{\tau s + 1}$, $u(t) = A \sin(\omega t) \Rightarrow U(s) = \frac{A\omega}{s^2 + \omega^2}$

- $Y(s) = \frac{KA\omega}{(\tau s + 1)(s^2 + \omega^2)}$, do PFE and inverse L.T. to obtain

$$\begin{aligned} y(t) &= \frac{KA\omega\tau}{1 + \omega^2\tau^2} e^{-t/\tau} + \frac{KA}{1 + \omega^2\tau^2} \sin(\omega t) - \frac{KA\omega\tau}{1 + \omega^2\tau^2} \cos(\omega t) \\ &= \frac{KA\omega\tau}{1 + \omega^2\tau^2} e^{-t/\tau} + \frac{KA}{\sqrt{1 + \omega^2\tau^2}} \sin(\omega t + \phi) \end{aligned}$$

- At sufficient large t , we have $y(t) = \hat{A} \sin(\omega t + \phi)$, where $\hat{A} = \frac{KA}{\sqrt{1 + \omega^2\tau^2}}$ and $\phi = -\arctan(\omega\tau)$
- For first-order system $G(s) = \frac{K}{\tau s + 1}$:

$$\text{Amplitude Ratio} = \frac{\hat{A}}{A} = \frac{K}{\sqrt{1 + \omega^2\tau^2}}, \quad \phi = -\arctan(\omega\tau)$$

Analysis: First-order lag

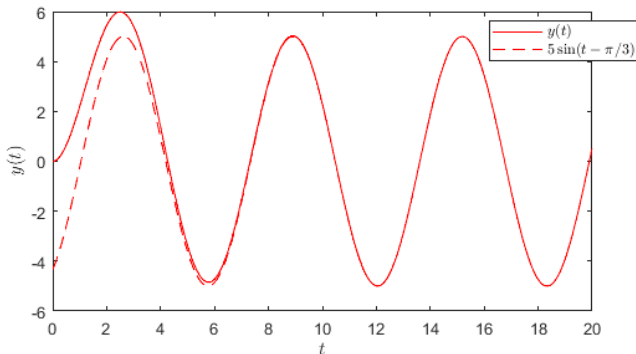
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- Amplitude ratio $\downarrow$ as $\omega \uparrow$: $G(s) = \frac{K}{\tau s + 1}$ is a “low-pass filter”
 - ▶ When $\omega \ll 1/\tau$, A.R. $\approx K$
 - ▶ When $\omega \gg 1/\tau$, A.R. $\approx K/\omega \tau \propto \omega^{-1}$
 - ▶ When $\omega = 1/\tau$ (called the cutoff frequency) A.R. $= K/\sqrt{2}$
- $\phi \downarrow$ from 0 to $-\pi/2$ as $\omega \uparrow$ from 0 to ∞ : $G(s) = \frac{K}{\tau s + 1}$ is a “lag”
 - ▶ When $\omega = 1/\tau$, $\phi = -\pi/4 (= -45^\circ)$.

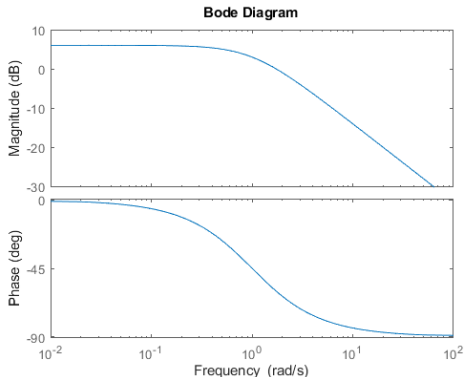
Example: First-order lag

- Exercise: $G(s) = 2/(\sqrt{3}s + 1)$, $u(t) = 5 \sin t$, then at sufficiently large t , $y(t) = ?$
 - $K = 2$, $\tau = \sqrt{3}$, $\omega = 1$. Hence A.R. = $2/\sqrt{1+3} = 1$,
 $\phi = -\arctan \sqrt{3} = -\pi/3$, $y(t) = 5 \sin(t - \pi/3)$.
- You can verify with a simulation.



Bode diagram

- Plot A.R. (in log scale) and ϕ (in linear scale) against ω (in log scale)
- Matlab: `[AR, phi] = bode(sys, omega)` or `bode(sys)`



- Decibels (dB)?? $\text{A.R. [in dB]} = 20 \log_{10} \text{A.R. [actual]}$
 - ▶ $\sqrt{10} = 10 \text{ dB}$, $2 \approx 6 \text{ dB}$, $\sqrt{2} \approx 3 \text{ dB}$, $1 = 0 \text{ dB}$, $0.5 \approx -6 \text{ dB}$...
 - ▶ The cutoff frequency is where the A.R. has dropped by 3 dB

General conclusion about frequency response

$G(s)$ at $s = i\omega$ determines frequency response!!

If $G(s)$ is a stable system, then $G(i\omega)$ in **polar coordinate** satisfies:

$$G(i\omega) = (\text{A.R.})e^{i\phi}.$$

In other words, at radial frequency ω ,

$$\text{A.R.} = |G(i\omega)|, \quad \phi = \angle G(i\omega)$$

- Example 1: First-order lag $G(s) = \frac{K}{\tau s + 1}$ ($K > 0$):

$$G(i\omega) = \frac{K}{1 + i\omega\tau}$$

$$|G(i\omega)| = \frac{|K|}{|1 + i\omega\tau|} = \frac{K}{\sqrt{1 + \omega^2\tau^2}} = \text{A.R.} \quad \checkmark$$

$$\angle G(i\omega) = \angle K - \angle(1 + i\omega\tau) = -\arctan(\omega\tau) = \phi \quad \checkmark$$

Example 2: Lead-lag

$$G(s) = \frac{1 + \tau_0 s}{1 + \tau_1 s} \quad s \stackrel{=}{\Rightarrow} i\omega \quad G(i\omega) = \frac{1 + i\omega\tau_0}{1 + i\omega\tau_1}$$

- Evaluate modulus and angle:

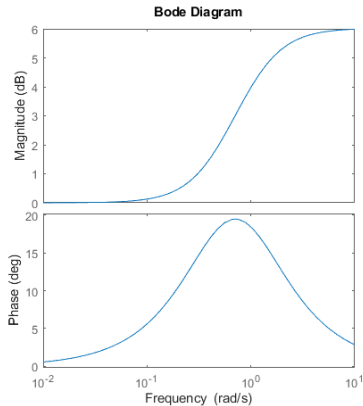
$$\text{A.R.} = |G(i\omega)| = \frac{|1 + i\omega\tau_0|}{|1 + i\omega\tau_1|} = \frac{\sqrt{1 + \omega^2\tau_0^2}}{\sqrt{1 + \omega^2\tau_1^2}}$$

$$\phi = \angle G(i\omega) = \angle(1 + i\omega\tau_0) - \angle(1 + i\omega\tau_1) = \arctan(\omega\tau_0) - \arctan(\omega\tau_1)$$

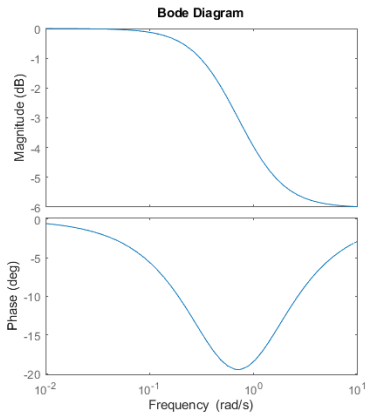
- When $\omega \rightarrow 0$: A.R. $\rightarrow 1$, $\phi \rightarrow 0$
- When $\omega \rightarrow \infty$: A.R. $\rightarrow \tau_0/\tau_1$, $\phi \rightarrow 0$
- If $\tau_0 > \tau_1$ (lead dominates), A.R. $\uparrow$, $\phi \geq 0$ and has a max.
- If $\tau_0 < \tau_1$ (lag dominates), A.R. $\downarrow$, $\phi \leq 0$ and has a min.

Example 2: Lead-lag

- Bode plot for $G(s) = \frac{2s+1}{s+1}$



- Bode plot for $G(s) = \frac{s+1}{2s+1}$



Example 3: Integrating system

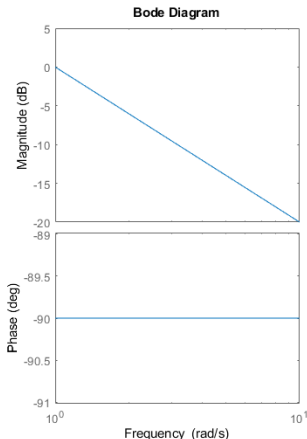
$$G(s) = \frac{K}{s} \quad (K > 0) \quad \xRightarrow{s=i\omega} G(i\omega) = \frac{K}{i\omega}$$

- Evaluate modulus and angle:

$$\text{A.R.} = |G(i\omega)| = \frac{|K|}{|i| \cdot |\omega|} = \frac{K}{\omega}$$

$$\phi = \angle G(i\omega) = \angle(K/\omega) - \angle i = -\pi/2$$

- A.R. $\propto \omega^{-1}$. The first plot of Bode is a line with slope -1 (in log-log)
- $\phi = -\pi/2$ for an integrator. The second plot of Bode is a constant



Example 4: Time delay

$$G(s) = e^{-\theta s} \ (\theta > 0) \quad \xrightarrow{s=i\omega} G(i\omega) = e^{-i\omega\theta}$$

- Evaluate modulus and angle:

$$\begin{aligned} \text{A.R.} &= |G(i\omega)| = |e^{-i\omega\theta}| = 1 \\ \phi &= \angle G(i\omega) = \angle e^{-i\omega\theta} = -\omega\theta \end{aligned}$$

- A.R. is a constant 1 (naturally!)
- ϕ is endlessly decreasing

